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The Middle Triassic magnetostratigraphy from the Peri-Tethys basin in Poland

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Abstract

The paper presents the first complete magnetic polarity scale obtained for the Middle Triassic of the northern Peri-Tethys area. The Roetian and Muschelkalk deposits of southern Poland (Upper Silesia, the Holy Cross Mountains) were studied paleomagnetically. The obtained paleopole 'A' (52°N, 143°E) fits well to the Middle Triassic (ca. 240 Ma) segment of the Stable European APWP. A set of various quality characteristic directions isolated in 106 samples was used for construction of the composite polarity scale. This Peritethyan scale has been tied with conodont zonation and referred to the scale erected earlier in the Tethyan realm. Comparison of both scales indicates that the whole Roetian succession in the southern Polish basin should be correlated with the latest Olenekian. The normal marine sedimentation ceased in the Polish basin as early as in the late Fassanian time. The obtained magnetostratigraphic scale combined with the sequence stratigraphic framework enables a reliable chronological correlation of the late Olenekian–Ladinian succession over the entire Peri-Tethys Basin. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: magnetostratigraphy; pole positions; Olenekian; Muschelkalk; Poland

1. Introduction

Detailed correlation of the Roetian–Muschelkalk succession of the northern Peri-Tethys (Germanic Basin) with its counterparts from the Tethys is problematic due to scarcity of index fossils common for the both basins. Especially the Roetian and Middle Muschelkalk are devoid of the index macro- and micro-fossils. Therefore the most difficult problem to solve is defining of the chronostratigraphic position of the Roetian–Muschelkalk boundary, i.e. the Lower–Middle Triassic boundary. Von Pia [1] assumed the lower boundary of the German Muschelkalk as concurrent with a beginning of the Anisian stage in the Alpine Triassic. This correlation has been accepted until 1974 when Kozur [2] placed the Olenekian–Anisian boundary within the Roetian sequence, at the base of the Myophoria Beds. In his recent work Kozur [3] has included the entire Roetian succession to the Anisian stage. The youngest position of Roetian–Muschelkalk boundary has been defined by Visscher et al. [4], which moved it up to the Pelsonian substage.

The magnetostratigraphy appears to be the

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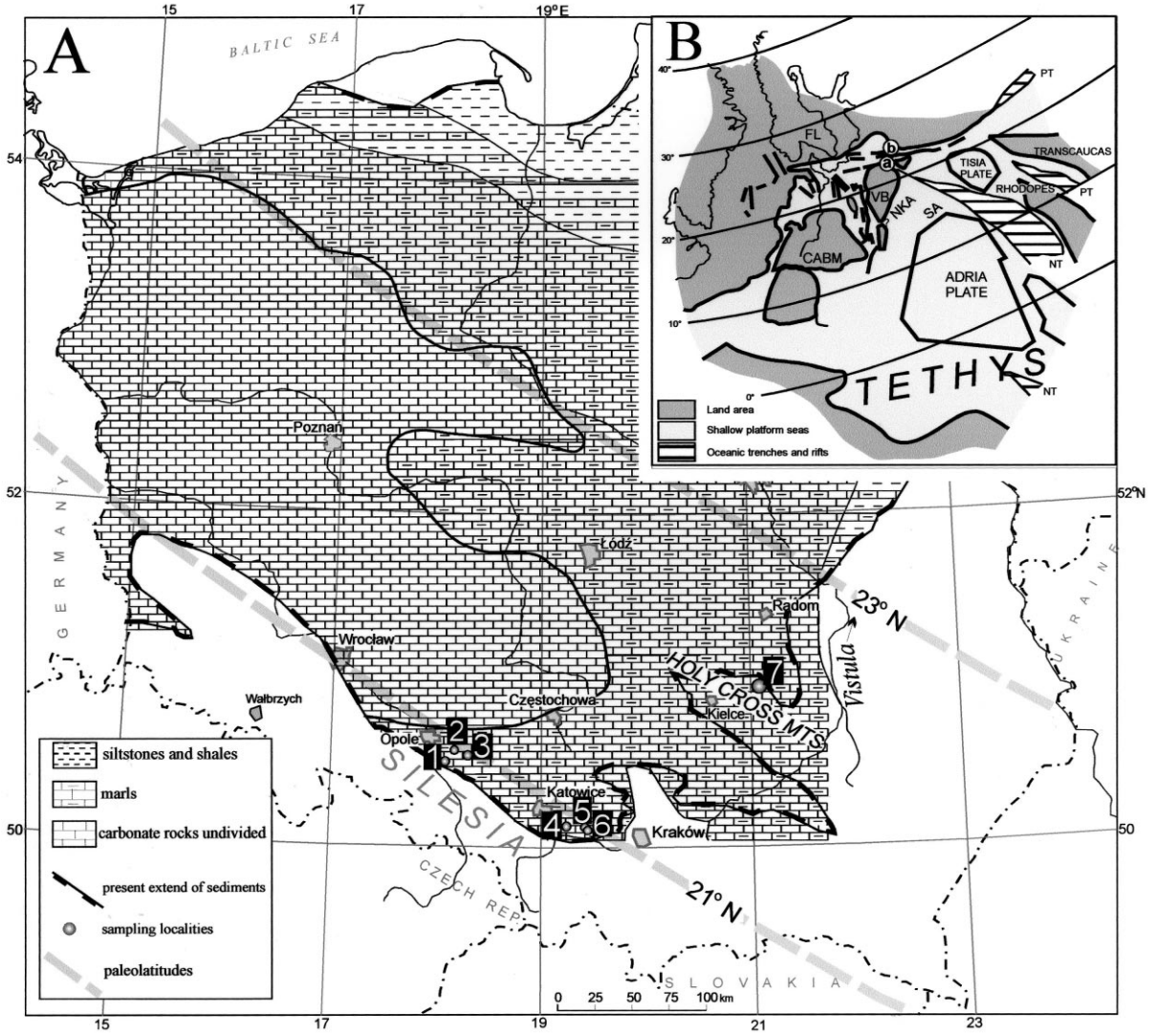


Fig. 1. (A) Generalized lithofacies map of the Lower Muschelkalk in Poland (after [38], modified) and paleomagnetic sampling sites (1: Tarnów Opolski, 2: Gogolin, 3: Strzelce Opolskie, 4: Libiąż, 5: Podstoki, 6: Płaza, 7: Nietulisko). (B) Paleogeographical map of the western Tethys area in the Anisian (after [9], modified). a: Upper Silesia, b: Holy Cross Mountains, FL: Fennoscandian Land, VB: Vindelician–Bohemian Land, CABM: Central, Armorican and Brabant Massifs, NKA: Northern Calcareus Alps, SA: Southern Alps.

most promising method to overcome the correlation hindrances. The presented work has been planned as continuation of the magnetostratigraphic examination carried out on the Permian–Lower Triassic succession from the Polish part of the Germanic Basin. This study enabled construction of the magnetostratigraphic scale for the Permian–Buntsandstein interval [5]. According to this scale the Griesbachian–Dienerian age of the Lower Buntsandstein and the lower part of Middle Buntsandstein has been confirmed. The entire Roetian (or at least its majority) has been correlated with the lowermost Anisian. However, the latter correlation was assumed as very hypothetical and conditional since the biostratigraphic guidelines were vague for this interval and the

polarity pattern in the overlying Muschelkalk sequence has not been recognized. Paleomagnetic investigation of the Roetian–Muschelkalk sediments was expected to approach the chronostratigraphic position of the discussed succession since in the Tethyan area progress has been made in calibration of the sequence of the Middle Triassic reversed and normal magnetozones against a detailed biostratigraphic zonation [6–8]. The presented magnetostratigraphic scale has been referred to the conodont biozonation erected for the Polish Muschelkalk by Trammer [9] and Zawidzka [10] and updated by Narkiewicz [11].

The next aim of this paper is to present a new Middle Triassic paleomagnetic pole that might be useful for definition of the Middle Triassic paleoposition of the European Plate. Until now only one good quality pole has been isolated from the Muschelkalk deposits [12].

2. Geology

2.1. General setting

The Triassic Germanic basin called also 'northern Peri-Tethys' occupied the northern margin of the Tethys Ocean (Fig. 1). This basin was partly open to the south and closed to the north. It was separated from the Tethys by elevated hercynian blocks cut by tectonically controlled depressions (gates). The gates provided communication between the northern Peri-Tethys and the Tethys Ocean, however they were opened in different times. The paleogeographical setting resulted in significant lithofacies diachroneity between the eastern (Polish) and the western (German) parts of the basin (see figure in the **EPSL Online Background Dataset**[1]). Normal marine and fossiliferous carbonates dominate in the Roetian and Lower Muschelkalk succession of southern Poland (Upper Silesia, Holy Cross Mountains). To the west these sediments are replaced gradually by deposits typical for more restricted and shallower environments, i.e. fossil poor, shallow water limestones and dolomites, sabkha evaporites and playa–mudflat siliciclastic deposits [13]. The circulation reversed in Ladinian time when the western communication dominated whereas the eastern and northern parts of the basin were uplifted and underwent emersion.

The facies diachroneity within the northern Peritethyan area makes it difficult to get an unequivocal basinwide correlation of the Triassic lithostratigraphic units on the one hand and hinders their reliable correlation with the Alpine Triassic succession on the other hand. The mentioned isolation and restricted circulation within the Peri-Tethys basin resulted in scarcity of index fossils useful for biostratigraphic correlation of the Germanic Triassic.

The first attempt to get an overregional biostratigraphic correlation of the lithostratigraphic units in the Germanic Basin has been carried out by Kozur [2], and based on integrated biostratigraphic studies of different fossil groups including the conodonts. The succeeding studies on conodont and foraminifera stratigraphy [9–11,14–16] enabled a relatively good biozonal subdivision of the Muschelkalk from the Pelsonian to Fassanian substages and enhanced correlation of these intervals with the Alpine Middle Triassic.

Recently, a sequence stratigraphic method has been employed to approach the problem of the regional lithofacies variation and for correlation of the sequences defined in the Tethys and Peri-Tethys basins [13,17]. A close number and timing of the depositional sequences featuring the both basins, argue for a dominant role of the eustatic controls upon the sedimentary processes both in the Tethys and its northern periphery [13].

2.2. The Upper Silesian section

The Roetian and Muschelkalk of Upper Silesia have been chosen for magnetostratigraphic study since the succession is complete and the rocks are quite well exposed with exception for the lower part of the Upper Muschelkalk. Moreover, the Silesian subbasin was situated within the transitional belt between the Tethys and the Germanic

[1] http://www.elsevier.nl/locate/epsl; mirror site: http://www.elsevier.com/locate/epsl

areas [17] which makes it easier to correlate both regions to each other.

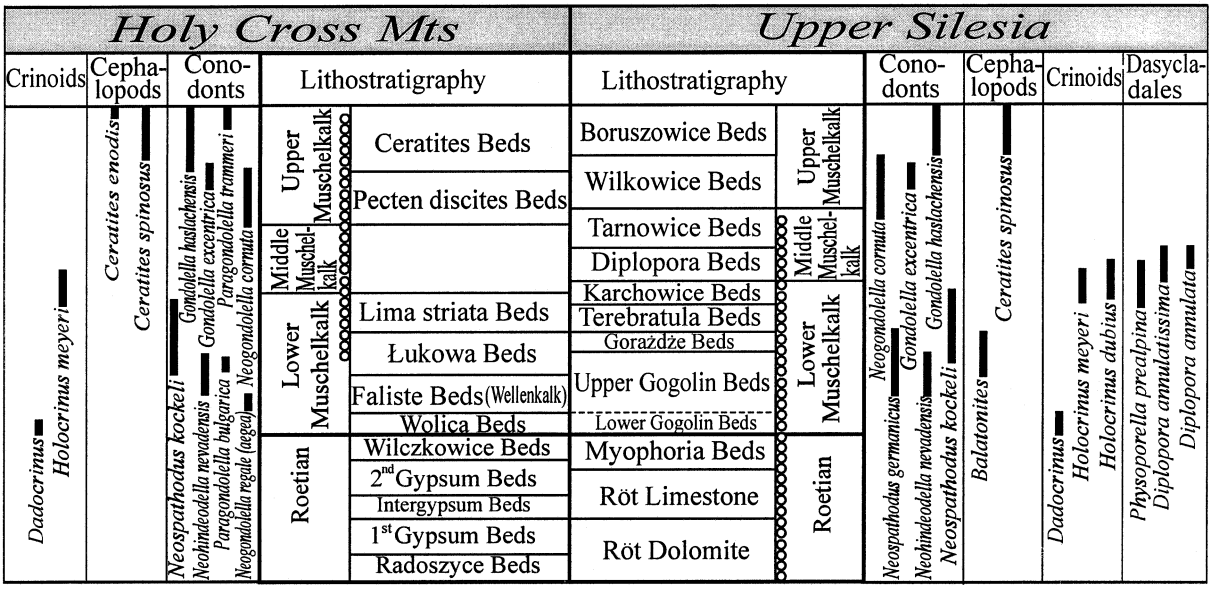
The Muschelkalk succession of Upper Silesia has a relatively well established conodont zonation founded by Zawidzka [10]. The Alpine ammonoids have long been reported from the Silesian Muschelkalk [18,19], but due to their sporadic occurrence they play a subordinate role in biostratigraphic zonation of the Muschelkalk. No age-diagnostic fossils have been so far found from the Roetian, lowermost part of the Lower Muschelkalk, Middle Muschelkalk and from the Lower Keuper interval.

The studied section starts with marls and oolitic limestones comprising the Roetian-indicative fossils: *Costatoria costata* and *Beneckeia tenuis* (Lower Roetian), and marly limestones with *Myophoria vulgaris* (Upper Roetian). The succeeding bioclastic limestones (Lower Gogolin Beds) comprising stenohaline *Dadocrinus* crinoids begin the Muschelkalk succession in the Upper Silesian Basin and may be correlated with the Scythian–Anisian transitional zone in the Alpine Triassic [20].

The first conodont taxa belonging to the *homeri–regale* composite zone (*N. germanicus*, N. cf. *homeri*) of the Aegean age occur ca. 30 m above the Roetian–Muschelkalk boundary, within the Upper Gogolin Beds [10]. The index conodont, *Neospathodus kockeli*, defining the beginning of the Pelsonian substage appears some 5 m below the Gogolin–Gorazdze Beds boundary and coincides with the appearance of the *Balatonites oltonis* ammonite, typical of the Pelsonian in the Alpine domain [21]. The next index conodont, *Paragondolella excelsa*, occurs first in the Karchowice Beds [10] and defines the lower limit of the *excelsa* assemblage zone. This limit is assumed by Zawidzka also as the Pelsonian–Illyrian boundary. Unfortunately the subsequent conodont-disadvantageous sedimentary environments resulted in stratigraphically barren zone encompassing the Diplopora Dolomite and the Lower Tarnowice Beds (Table 1). Therefore the boundary position is not definite. According to the occurrence of dasycladales algae Kotanski [22] has placed the boundary higher, within the Diplopora Dolomite.

Stratigraphic zonation of the Upper Muschelkalk constructed by Kozur [23] for the German Muschelkalk, is based on phylogenetic stages of the genus *Gondolella* and consists of seven cono-

Table 1
Roetian–Muschelkalk lithostratigraphy and selected fossil extents in the Holy Cross Mountains and Upper Silesia



Paleomagnetic sampling intervals are marked by circles.

dont zones. The first, after the mentioned intermission, index conodont assemblage (including *Neogondolella mombergensis*) appears only in the upper Tarnowice Beds and defines the 'first conodont zone' of the Upper Muschelkalk. The subsequent 'second conodont zone' is defined by the occurrence of *Neogondolella cornuta*, *Neogondolella constricta* and *Neogondolella excentrica* taxa. Zawidzka [10] assumed the first appearance of the latter taxon as an indicator of the Anisian–Ladinian boundary which lies within the Wilkowice Beds. The third and fourth conodont zones are determined by first occurrences of *Neogondolella media* and *Paragondonella haslachensis*, respectively. The last-named zone, indicative for the Late Fassanian and earliest Longobardian, encompasses the entire Boruszowice Beds [10]. The subsequent conodont zones 5–7 are absent in the Silesian Muschelkalk since the normal marine conditions were replaced by brackish, clastic-diluted sedimentation, typical of the Lower Keuper environments.

2.3. The northern Holy Cross Mountains section

As indicated by paleontological data [24,25] the Muschelkalk succession in the northern Holy Cross Mountains commences only with the Pelsonian sediments overlapping the Lower Roetian clastics. Since the succeeding deposits represent almost complete late Pelsonian–Fassanian succession, this profile has been chosen for construction of the composed magnetostratigraphic scale of the Peritethyan Middle Triassic.

The generally accepted lithostratigraphic division of the Triassic succession of the Holy Cross Mountains has been erected by Senkowiczowa [26] and this scheme is applied also in our work (Table 1). The first age-diagnostic conodonts, *Gondolella aegea* (= *Gondolella regale*) have been found by Trammer [9] within the Wellenkalk Beds. The occurrence range of this taxon defines the Aegean age of the Wellenkalk Beds. The subsequent taxon *Paragondolella bulgarica* which occurs in the lower part of the Lukowa Beds defines the Bithynian age of this interval. The next index conodont taxon, *N. kockeli*, accompanied by *Paragondolella navicula*, *P. excelsa* and *Paragondolella bifurcata* which occur in the upper part of the Lukowa Beds and in the Lima striata Beds define the Pelsonian age of this interval. The last named three taxa occur in the northern Holy Cross Mountains some 1–1.5 m above the Roetian–Muschelkalk boundary [24]. This indicates that the Muschelkalk transgression reached the northern Holy Cross Mountains area as late as in Pelsonian time, i.e. during the maximum flooding phase of the Lower Muschelkalk. This inference is confirmed by the coexisting crinoid fauna of the *Silesiacrinus silesiacus* zone (Hagdorn in [25]), which comprises in Upper Silesia the Karchowice and Diplopora Beds, representing the Mid–Late Pelsonian–Illyrian interval [20]. The Middle Muschelkalk facies featured by evaporites and emersion events are devoid of index fossils. They appear only in the Upper Muschelkalk and comprise species typical for its lower part (Pecten discites Beds), i.e. *Gondolella excentrica*, *Gondolella cornuta* and *Gondolella mombergensis* [24]. These taxa represent the 1st and 2nd conodont zones (see above). The uppermost part of the Muschelkalk in the studied section comprises *Ceratites spinosus* cephalopod remnants (Table 1; Hagdorn, in [25]) and *G. mombergensis* media conodonts defining the third conodont zone [24]. Since *Gondolella haslachensis* has not been found in the Muschelkalk deposits of the northern Holy Cross Mountains section, one may assume that the normal marine deposition terminated here one conodont zone earlier as in the southern Holy Cross Mountains area and in Upper Silesia, and four zones earlier as in the German part of the Muschelkalk Basin.

3. Sampling localities and paleomagnetic procedure

A total of 171 drill samples for paleomagnetic studies were collected from 132 beds exposed in six quarries and two ravines situated in the Silesia and Holy Cross Mountains (Fig. 1). Up to three specimens were cut from each sample. A set of samples taken from the same bed was considered as one magnetostratigraphic sample. The group of specimens obtained from one independently oriented sample was regarded as one sample used for

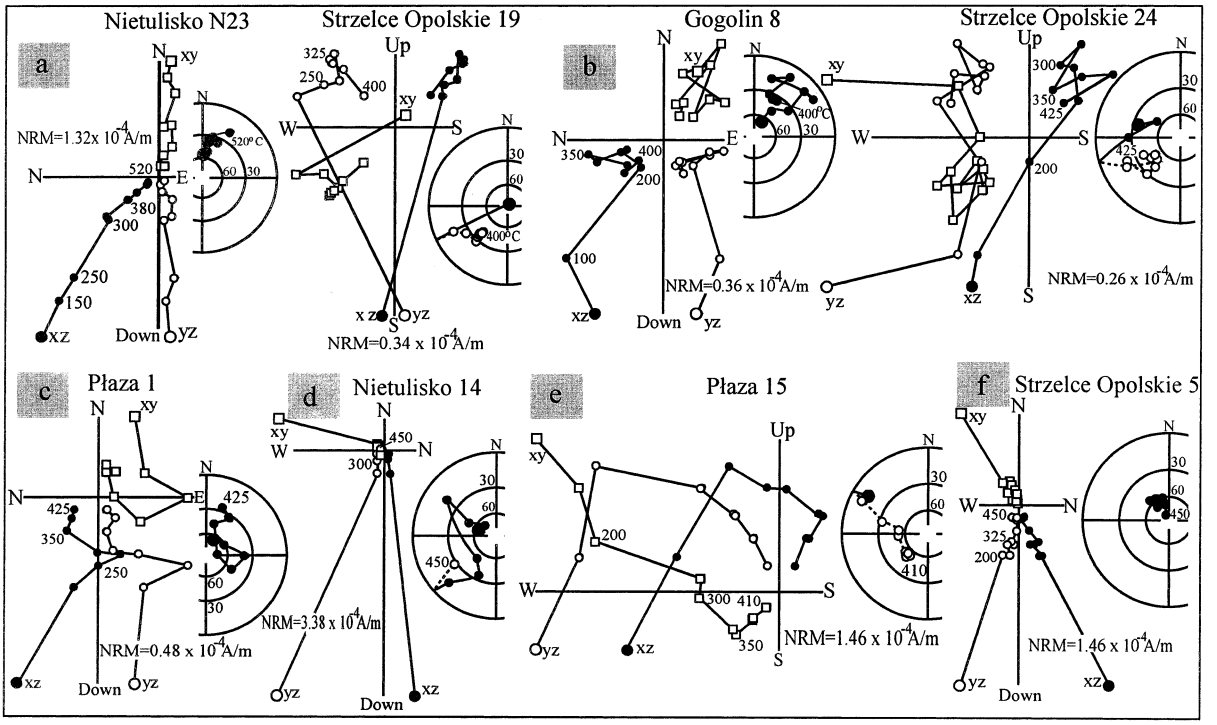


Fig. 2. Orthogonal demagnetograms and stereographic projections of NRM demagnetization directional tracks prepared for representative samples of Roetian–Muschelkalk rocks from the Holy Cross Mountains and Upper Silesia. (a) Samples containing well-defined (see text) component A. (b) Samples with component A oscillating near the expected Middle Triassic position. (c) Partly remagnetized sample that showed the expected Middle Triassic direction A at two last demagnetizations levels. (d) Sample containing component A that is strongly overlapped by a steep component of presumed Cenozoic age. The expected Middle Triassic position of paleodirection has been attained at the last demagnetization level only. (e) Sample with well-defined secondary component B of Late Triassic age. (f) Totally remagnetized sample containing Cenozoic component only. In the last group all data are presented after bedding tilt correction. In steroeplots, open (closed) symbols denote upward (downward) pointing inclinations.

paleopole calculation. In all localities the studied rocks were characterized by a very small dip that did not exceed 7°.

The natural remanent magnetization (NRM) of the specimens was measured using a JR-5 spinner magnetometer with a noise level about 0.3×10^{-5} A/m. Some pilot specimens were subjected to alternating field (AF) and thermal demagnetization experiments. Since the AF method was not effective the whole sample set has been subjected to the stepwise thermal demagnetization in a μ-metal shielded oven, which reduced the ambient field close to a few nT. After each thermal demagnetization level a magnetic susceptibility signal was monitored. Least-square line fit methods, as presented by Kirschvink [27], were used to calculate the components of NRM and their unblocking temperature spectra. Thermomagnetic analysis and analysis of isothermal remanent magnetization (IRM) acquisition were used to determine the nature of magnetic carriers.

4. Results of paleomagnetic analysis

About 50% of the specimens revealed NRM intensities lower than 0.5×10^{-4} A/m. The NRM intensities in 38% of the specimen collection dropped down below 0.7×10^{-4} A/m at demagnetization temperatures higher than 200°C. 28 samples taken from 26 beds yielded well-defined, antipodal directions of presumed Middle Triassic age,

based on at least three subtracted vectors and a fitted-line with an angular standard deviation not higher than 5° (Figs. 2a and 3). They were used for Middle Triassic paleopole calculations (Table 2, component A). During thermal experiment 38 samples coming from 31 beds yielded directions that oscillated near the expected Middle Triassic position (Fig. 2b). These small oscillations were connected with measurement errors caused by extremely low intensities. They did not allow to distinguish adequate fitted-line directions. In the other 41 samples from 35 beds the expected Middle Triassic component A was reached at the two last demagnetization levels (Fig. 2c). It was partly covered by a component with a steep inclination that corresponds to the local present-day field direction. 23 samples from 14 beds showed the expected Triassic position of paleodirection only at the last demagnetization level (Fig. 2d) that usually preceded an abrupt increase of magnetic susceptibility. These samples contain component A strongly overlapped by a steep component referring to the local present-day field direction. The expected Middle Triassic position of paleodirection has been attained at the last demagnetization level only. The above presented subdivision of demagnetization resulting into four paleomagnetic behaviors was used for determination of four categories of reliability of magnetic polarity. Obviously, the most reliable (first category) polarities were determined on the base of well-defined fitted-line directions. The last (fourth) category of polarity determination was based on the samples showing expected paleodirections at the last demagnetization level only. It should be stressed that the reliability grading of magnetostratigraphic data from weakly magnetized beds has been widely applied by other authors (e.g. [28]).

Ten samples from the upper part of the Lower Gogolin Beds (Plaza quarry) revealed the presence of well-defined component with characteristic inclination of about 15° higher than expected for the Middle Triassic (Fig. 2e). Because of this and because of different magnetic mineralogy (see Section 5) this remanence was regarded as secondary in origin (Table 2, component B). Nineteen samples taken from 11 beds were totally remagnetized in the present field direction (Fig. 2f).

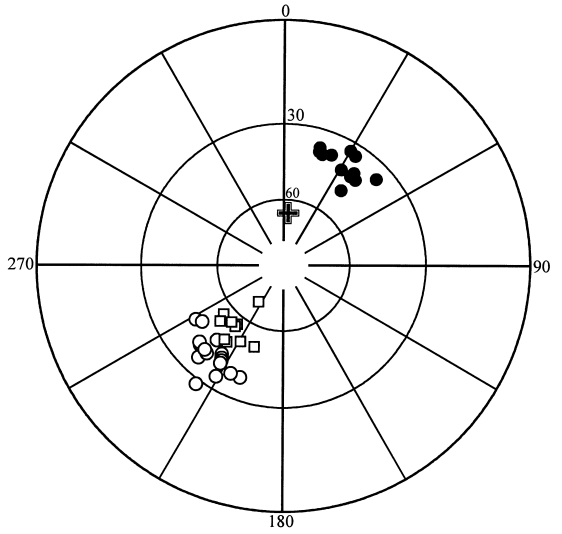


Fig. 3. Stereographic projection of the primary Muschelkalk (circles) and secondary Late Triassic (squares) line fitted directions from the Upper Silesia and Holy Cross Mountains (for summary of statistics see Table 2). Cross shows the local present-day field direction. All directions are presented after bedding tilt correction. Open (closed) symbols denote upward (downward) pointing inclinations.

5. Magnetic carriers

Magnetic minerals from the Muschelkalk carbonates of the Silesia region have been already analyzed by Symons et al. [29]. According to the results of that study Triassic remanence resides in single–pseudosingle domain magnetite in limestone and is probably primary, and in single–pseudosingle domain magnetite and hematite in early dolomite and is probably diagenetic [29]. Indeed, the IRM acquisition and thermomagnetic curves prepared for the Silesian samples with the Muschelkalk remanence (component 'A') show the presence of low coercivity mineral with maximum unblocking temperatures not higher than 580°C (Fig. 4; sample PL5e and SO24). Different petromagnetic behavior displays samples from the Holy Cross Mountains that also contain the Muschelkalk component. Apart from magnetite it reveals the presence of goethite and hematite (Fig. 4; sample NT25). Samples with secondary com-

ponent 'B' isolated in the Plaza section (Gogolin Beds) contain a high coercivity mineral that was demagnetized at temperatures higher than 600°C (Fig. 4; sample PL12). These properties indicate the presence of hematite. Samples totally remagnetized in the present field direction contain mainly goethite (Fig. 4; sample SO27).

6. Paleomagnetic poles

Paleopoles were computed only for localities containing more than five independently oriented samples with well-defined fitted-line directions. These paleopoles and the mean Muschelkalk paleopole are listed in Table 2. The mean Muschelkalk paleopole (labeled 'A') and the paleopole from the remagnetized zone detected in the Plaza section (labeled 'B') have been compared with the apparent polar wander path (APWP) characteristic for Stable Europe. This APWP was constructed using only good quality ($Q > 4$) poles. The Early Triassic poles were taken from the list of Van der Voo [30]. They were obtained in the cratonic area of Europe (i.e. East European Platform). The six best quality Middle Triassic–Early Jurassic poles were determined in NW France and SW Germany [12,31]. The calculation and smoothing of the apparent wander path was performed using the GMAP plotting package [32]. The smoothing procedure involves the spherical spline method and the following splining parameters have been applied: tension factor = 200, time resolution = 2.5 Ma. It is clearly visible that the Muschelkalk paleopole 'A' fits well to the Middle Triassic (ca. 240 Ma) segment of the Stable European APWP (Fig. 5). Paleopole 'B' from Plaza is shifted towards the Late Triassic part of this APWP, similarly as the paleopole obtained earlier from the Silesian Muschelkalk by Symons et al. [29].

The paleolatitudinal data are important in paleoecological and paleofacial models. According to paleoenvironmental maps of Dercourt et al.

Table 2
Summary of statistics of paleodirections isolated in the Muschelkalk of the Polish basin and selected Late Permian–Middle Triassic poles used for construction of the Stable European APWP (Fig. 5).

Locality	Group	Comp.	N	D	I	$\alpha 95$	K	Lat.	Long.	dp	dm	Pol.	Plat.
Płaza	Lower	A	9	210	−37	5.3	95.8	52°N	149°E	4	6	Mix.	21°N
(19.5°E, 50.1°N)	Muschelkalk (55°/3°)*	B	10	217	−54	5.1	90.7	59°N	126°E	5	7	R	34°N
Strzelce Opolskie (18.2°E, 50.5°N)	Lower/Middle Muschelkalk (42°/4°)*	A	12	217	−41	4.6	91.1	51°N	138°E	3	6	Mix.	23°N
Nietulisko (21.1°E, 50.9°N)	Middle/Late Muschelkalk (223°/3°)[a]	A	7	216	−42	7.0	74.9	51°N	142°E	5	9	Mix.	24°N
Polish basin	Muschelkalk	A	28	214	−40	3.0	84.2	51°N	143°E	2	4	43% N, 57% R	23°N
(summary)			3[a]	214	−40	6.0	421.5	52°N	143°E	4	7		
1. France,	Illyrian/Fassanian carbonates					3		53°N	141°E				
Ukraine	Serebryansk, Donbas (Early Triassic)					12		56°N	146°E				
Russia	Varieg. Suite, Urals (Early Triassic)					4		51°N	151°E				
2. Russia	Romasska Redbeds (Early Triassic)					4		52°N	165°E				
Russia	Veltugian Red Clays					4		53°N	158°E				
Russia	Upper Tatarian Redbeds					4		51°N	166°E				

Comp. = name of component; N = number of independently oriented samples ([a]number of localities); D = declination; I = inclination; $\alpha 95$, K = radius of the cone of 95% confidence and precision parameter [43]; Lat. = latitude of north paleomagnetic pole; Long. = longitude of north paleomagnetic pole; dp = error of the distance between site and paleopole; dm = paleolongitude error; Pol. = polarity (Mix. = mixed, N = normal, R = reversed); Plat. = paleolatitude calculated for Plaza locality. *Mean azimuth and magnitude of dip. All data are presented after bedding correction. 1 = Middle Triassic pole of Theveniaut et al. [12], 2 = Late Permian–Early Triassic poles listed by Van der Voo [30].

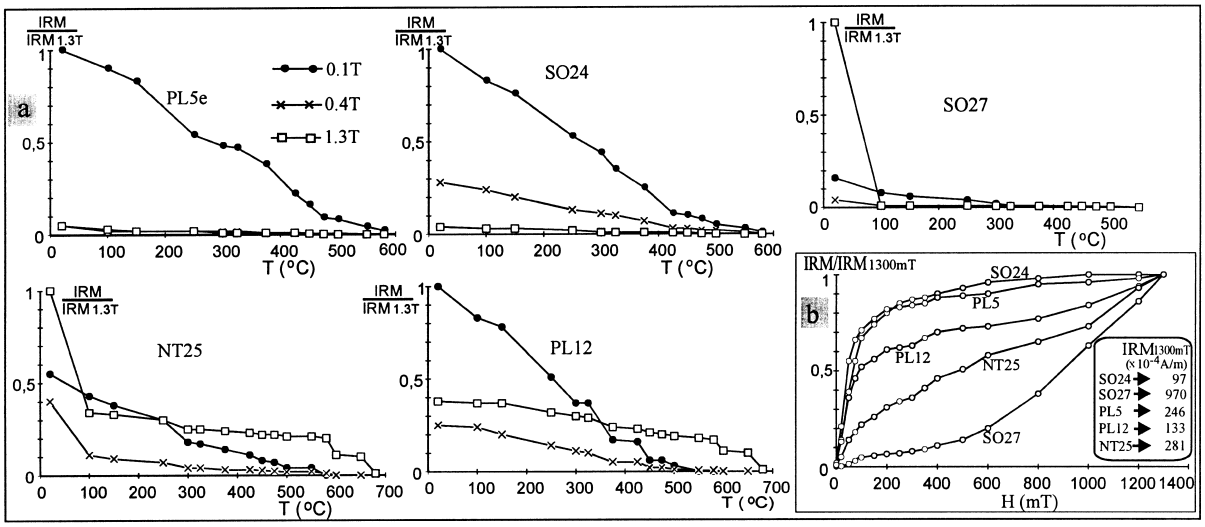


Fig. 4. (a) Thermal demagnetization of orthogonal-axis IRM curves obtained for selected Roetian–Muschelkalk samples from the Upper Silesia (PL5e, PL12, SO24, SO27) and Holy Cross Mountains (NT25). (b) IRM acquisition curves prepared for the same sample set.

[33] in the Middle Triassic time Silesia was located around 32°N. The Muschelkalk inclinations obtained in this paper were derived mainly from the Anisian samples. These inclinations correspond to 21–24° of paleolatitude (Table 2).

7. Magnetic stratigraphy and correlation

7.1. Upper Silesia

A very good (bed to bed) internal stratigraphic correlation of Silesian outcrops allows to present the paleomagnetic data in the one diagram comprising the Roetian–Middle Muschelkalk strata (Fig. 6). Magnetostratigraphic subdivision of these rocks is based on the results of various reliability. In the case where well-defined fitted-line directions were not obtained the declination and inclination diagrams were constructed by the means of directions obtained at the last demagnetization level. However, no magnetozone was distinguished using poor quality (third and fourth category) samples only. The magnetozones were numbered if the same polarity record was noted in more than one sample. In this way seven normal and seven reversed magnetozones were defined in the Silesian profile. The normal polarity record seems to be predominant in the lower part of the Roetian and lowermost Muschelkalk. Reversed directions of magnetization occur especially in the Pelsonian segment of the section.

A certain break in the magnetic polarity scheme, related to the Late Triassic remagnetization, occurs in the middle part of Gogolin Beds (in Fig. 6 marked by gray belts). The origin of Late Triassic secondary component B is not evident. One possibility is to link remagnetization with the fluid–rock interaction. In the Gogolin Beds authigenic potassium feldspar was found [34]. It has been ascribed to burial diagenesis proceeded without contact with seawater. The K-feldspar has been earlier defined as a tracer of paleofluid flow in carbonates [35,36] related to their remagnetization [37].

Volume magnetic susceptibility of the Muschelkalk carbonates from Silesia varies between -68 and 287×10^{-7} SI units. However these variations are not accidental. It is clearly visible that in several places maximum values of magnetic susceptibility are in agreement with the boundaries of depositional sequences (Fig. 6). Hence, magnetic susceptibility graphs can be used as a tool of stratigraphic correlation of these rocks.

7.2. The Holy Cross Mountains

The Nietulisko section is composed of Pelsonian–Early Fassanian carbonates with a certain break near the Pelsonian–Illyrian boundary [13]. A continuous magnetostratigraphic graph was prepared only for the upper part of the section where good quality Middle Triassic fitted-line directions were obtained (Fig. 7 and Table 2). Its lower part yielded hardly ever six dispersed points where polarities were defined. Three normal and four reversed magnetozones were established in the Illyrian–Fassanian part of the studied section. Normal polarity record seems to be predominant in the upper part of the Illyrian and lowermost part of Fassanian strata.

7.3. The composite polarity scale

The composite polarity scale (Fig. 8) has been constructed using a set of characteristic directions isolated in 106 samples. Magnetic polarity pattern in the Roetian and Lower Muschelkalk was defined on the basis of the results from Silesia. The Upper Muschelkalk part of the scale was adopted from the Nietulisko section. Magnetostratigraphic data from Nietulisko and Libiąż sites were used for construction of the Middle Muschelkalk segment. Correlation of the Muschelkalk Beds from the Holy Cross Mountains and Silesia was performed according to conodont biozonation [9,10] and the sequence stratigraphic procedure [13].

Within the entire Roetian–Muschelkalk se-

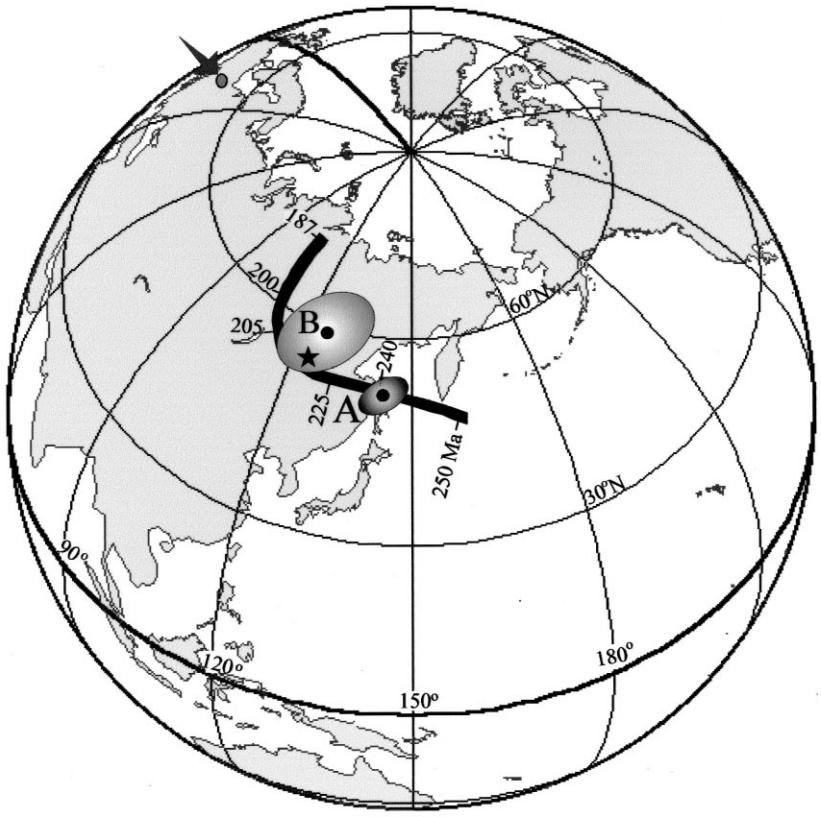


Fig. 5. Triassic–Early Jurassic APWP for European Plate (for construction details see text) and the Middle Triassic paleopole (labeled A) from this paper. Paleopole calculated for the secondary component B, and paleopole obtained earlier [29] from the Muschelkalk of Silesia (marked by star) are also presented. The study area is indicated by the arrow. The ages were adopted from the time calibration of Menning [39] and Mundil et al. [40] (Early–Middle Triassic), Kent and Olsen [41] (Late Triassic), and Palmer [42] (Early Jurassic). They are shown in Ma.

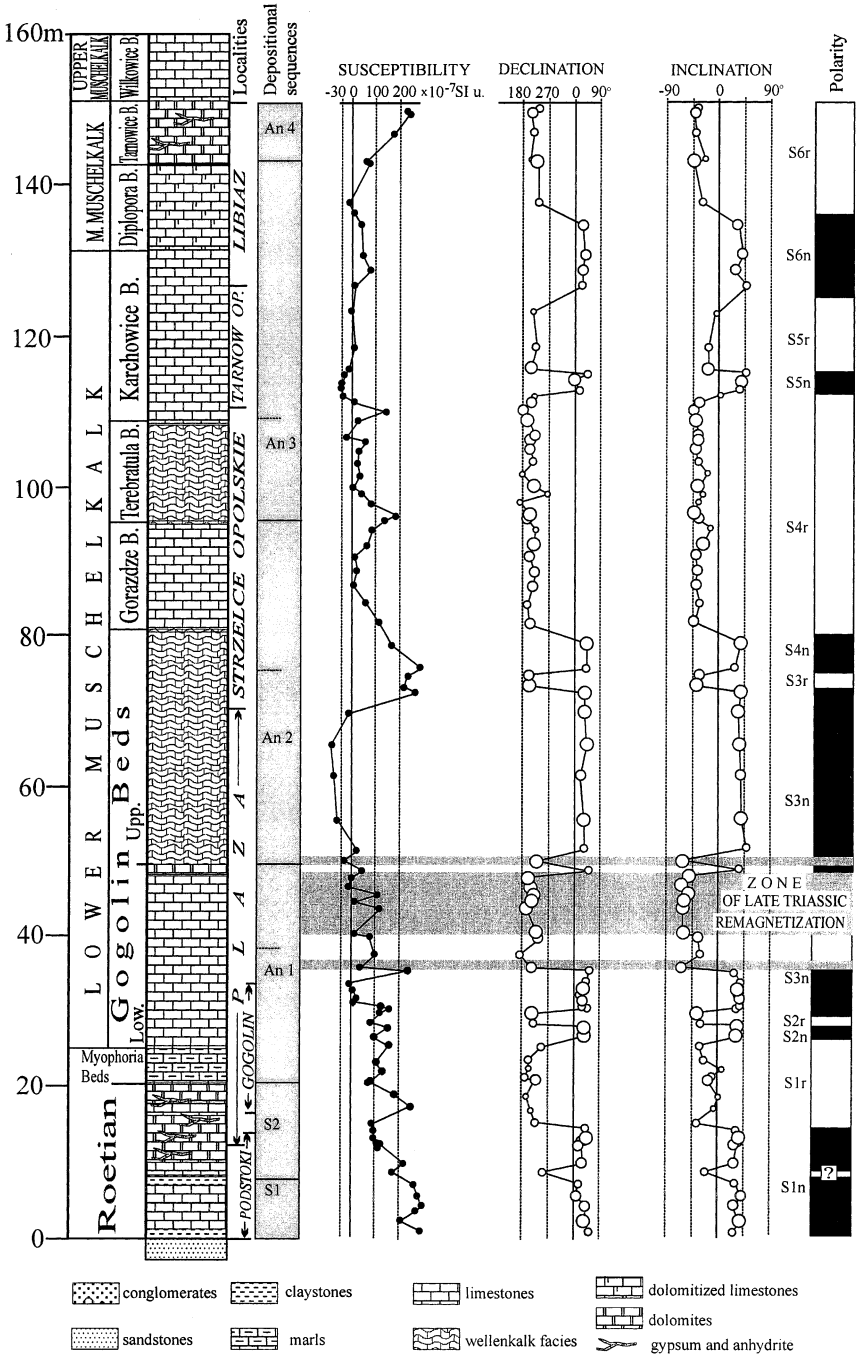


Fig. 6. Lithostratigraphy and sequence stratigraphy of the Roetian–Muschelkalk deposits from Upper Silesia and the results of their magnetostratigraphic investigations. On the right of lithostratigraphic column, the plots of magnetic susceptibility, characteristic declination, characteristic inclination and magnetic polarities are presented (black belt = normal polarity record, white belt = reverse polarity record). In the declination and inclination plots the biggest circles represent the first (best) category of data (see text). The two next smaller size circles are characteristic for second and third category data. Fourth category data yielded inclinations and declinations marked by the smallest circles. The polarity column is divided into numbered magnetozones.

quence nine normal and nine reverse polarity zones have been found. The magnetic polarity pattern noted in the Roetian sediments from Silesia is the same as in the Roetian section drilled in the Polish Lowland, where a normal polarity record was characteristic for the Lower Roetian, in contrast to its upper part that was magnetized reversely [5].

7.4. Magnetostratigraphic correlation with the Tethyan realm

The index conodonts common for the Holy Cross Mountains, Upper Silesia and the Alpine area were fundamental for construction of the Middle Triassic biostratigraphic framework in southern Poland (Table 1) and for its correlation with the Tethys section (Fig. 8). Other fossils

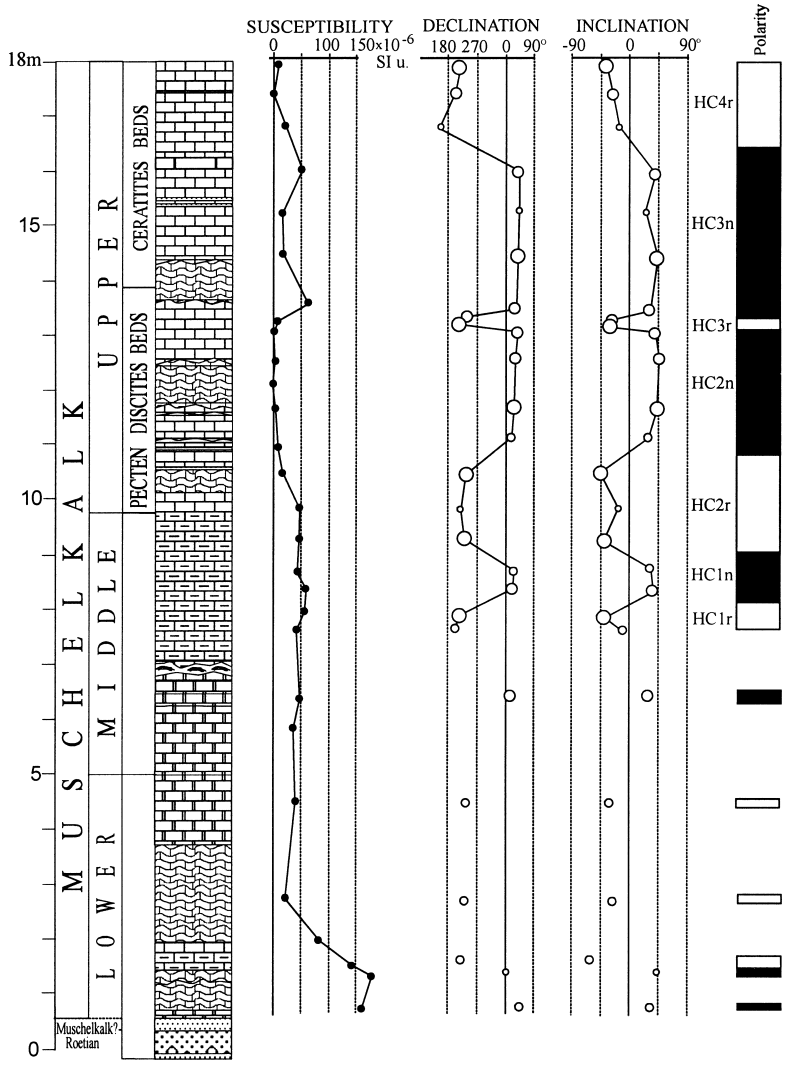


Fig. 7. Lithostratigraphy of Muschelkalk deposits from the Holy Cross Mountains (Nietulisko section) and the results of their magnetostratigraphic investigations. On the right of the lithostratigraphic column, the plots of magnetic susceptibility, characteristic declination, characteristic inclination and magnetic polarities are presented.

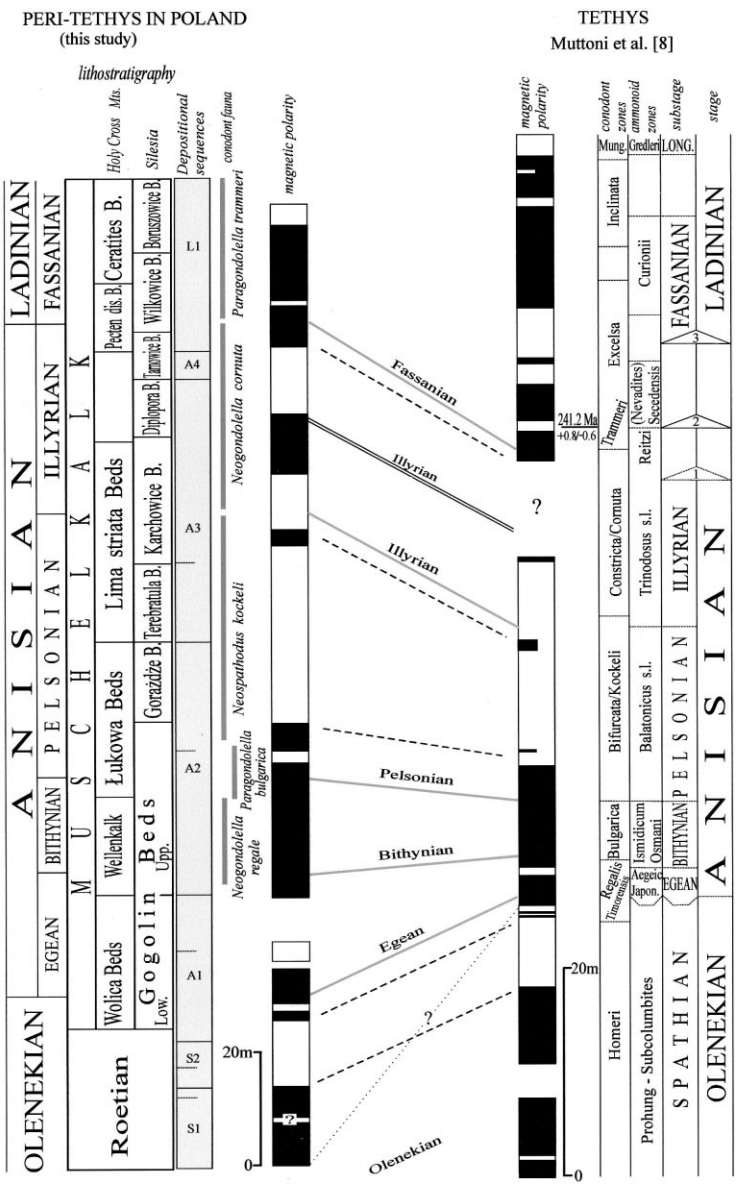


Fig. 8. Composite Roetian–Muschelkalk polarity scale for the Peri-Tethys basin in Poland and its correlation with polarity scale composed [8] for western Tethys. Broken lines demonstrate magnetostratigraphic correlation. Solid lines shows boundaries of substages. Dotted line with a question mark presents the hypothetical correlation (rejected in this paper) that would include the whole Roetian into the Anisian. Double line signals the Pelsonian–Illyrian boundary placed according to the occurrence of dasycladales algae [22].

(crinoids, cephalopods, dasycladales) appeared as useful aid-tools of the chronostratigraphic correlation between the both basins. The biostratigraphic tools contrived an adequate background for correlation of the Peritethyan Middle Triassic magnetostratigraphic scale with the Tethyan scale by Muttoni et al. [8]. The resulted magnetostratigraphic correlation indicates that the entire Roetian succession from the southern Poland represents the Olenekian age. Also the Muschelkalk

facies onset is coincidental with the latest Olenekian. As already mentioned, Kozur [3] has ascribed the entire Roetian succession to the Anisian stage based mainly on a holothurian fauna that can not be indexed with fossils since its real stratigraphic range is unknown. Regarding Kozur's concept as correct one should assume the Aegean age of the whole Roetian (cf. the correlation line marked with '?' in Fig. 8). However, it would be very difficult to interpret the presence of only one reverse zone in the Aegean and Bithynian substages within the Tethys profile and as much as three reversed zones in the Roetian–Lower Muschelkalk of the Peritethys basin. The presented magnetostratigraphic data support the older concepts of the correlation of the Triassic from the Germanic and Alpine areas, assuming the beginning of the Muschelkalk as nearly concurrent with the Scythian–Anisian boundary (e.g. [1]).

The end of the marine sedimentation of the Muschelkalk facies in the Polish basin took place in Fassanian time, i.e. much earlier than in the German basin. The earlier constructed sequence stratigraphic framework throughout the entire Peri-Tethys Basin [13,17] coupled with the constructed magnetostratigraphic scheme gives a way for chronostratigraphic correlation of the most important events in the basin history.

8. Conclusions

Eighteen biostratigraphically calibrated magnetozones were recognized in the Roetian–Muschelkalk sequence from the Peri-Tethys basin in Poland. The correlation of established magnetic polarity scale with the scale erected earlier in the Tethyan realm indicates that the whole Roetian succession in the southern Polish basin should be included to the Olenekian stage. The normal marine sedimentation ceased in the Polish basin as early as in the Late Fassanian time. Our studies show that the studied Muschelkalk sequence was deposited during the latest Olenekian–Early Fassanian time. The primary paleomagnetic directions from examined rocks gave a pole that fits well to the Middle Triassic segment of the Stable European APWP. The Muschelkalk inclinations obtained in this paper corresponds to 21–24° of the paleolatitude of the studied area.

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